Day 57 - AI System Stability and Refinement

Stable AI system | Debugging report | Final refined modules

Objective and scope

Day 57 consolidates the project work on scoring consistency, conversation flow, resume data handling, error reporting, and reliable outputs. The changes refine the existing Zecpath AI prototype and its local validation flows.

Deliverable summary

Deliverable	Completed work
Stable AI system	Aligned score ranges and shared scoring rules across pipeline stages. Adapted parsed and legacy resumes to the ATS schema, applied demographic masking to the scoring copy, and kept pending follow-up answers attached to the question that prompted them.
Debugging report	Added stage-specific resume input, extraction, parsing, and output errors. Implemented structured education and experience extraction, handled overlapping work periods, and hardened malformed or incomplete inputs.
Final refined modules	Standardized score response fields and JSON persistence. ATS components remain finite; saved JSON rejects non-standard numeric values and uses atomic replacement to preserve an existing file when a write fails.

System path and checkpoints

Stage	Checkpoint
Resume intake	Validate the PDF or DOCX path, extract readable text, and produce structured sections.
Scoring input	Convert parser or legacy data to the ATS schema and apply demographic masking to a scoring copy.
Screening flow	Keep dynamic follow-up answers connected to their prompt and apply bounded quality retries.
Decision outputs	Use stable score fields and strict, atomic JSON persistence for score and interview outputs.

Debugging report

Observed issue	Resolution
Score scales and role weights differed between stages.	Centralized unified and legacy role weights in shared configuration and kept screening scores on a consistent 0-100 scale.
A follow-up response could be stored against the next main question.	Stored the active dynamic follow-up prompt in conversation state and route the next response to that pending prompt, including after silence retry.
Parser output and ATS scoring expected different resume shapes.	Added a schema adapter for parser and legacy resumes; ensured masked resume data is used for skill, experience, education, and semantic scoring.
Resume failures exposed inconsistent low-level errors.	Added shared path validation and typed input, extraction, parsing, and output errors; reject empty extracted text and preserve dotted filenames.
Invalid scores or failed writes could produce unsafe output.	Normalize non-finite ATS component scores and save score, recommendation, and interview JSON through strict atomic writes.

Edge-case validation and refined modules

The final validation focused on boundary inputs that can otherwise cause a crash, misleading score, mismatched answer, or unstable saved output.

Edge cases checked

Case	Verified behavior
Invalid audio confidence or noise metadata	Malformed, boolean, non-finite, and out-of-range values trigger the applicable quality retry instead of raising an exception or silently passing.
Corrupt retry counter	Invalid retry state falls back safely rather than creating an unbounded retry.
Malformed job-description fields	Wrong-type role, skills, experience, and education fields are normalized before scoring; a single skill string remains one skill.
Resume experience values	Non-finite and negative experience totals normalize safely; malformed dates are skipped and future-dated work is capped at the current date.
Unordered set inputs	Set-valued skills and project fields are sorted to keep normalized output stable.
Invalid JSON serialization	Strict JSON rejects NaN, and an unsuccessful write leaves the previous file intact.

Final refined modules

Area	Refined modules
Scoring and resume adaptation	scoring/candidate_matching_service.py; scoring/resume_schema_adapter.py; scoring/ats_scoring_engine.py; scoring/unified_scoring_engine.py
Conversation and response quality	screening_ai/conversation_flow.py; screening_ai/response_quality_checker.py; screening_ai/edge_case_handler.py
Resume processing	parsers/resume_errors.py; parsers/resume_engine.py; parsers/resume_section_parser.py; parsers/resume_experience_extractor.py; parsers/resume_education_extractor.py
Output persistence and regression checks	utils/json_output.py; scoring/unified_candidate_score_generator.py; scoring/final_recommendation_generator.py; interview_ai/hr/interview_summary_pipeline.py; tests/test_day57_api_output_stability.py; tests/test_day57_edge_case_validation.py

Validation results

44 focused regression tests passed across the Day 57 API-output and edge-case checks, conversation flows, resume parsing, data adaptation, and candidate matching.

Status: the Day 57 deliverables are complete and the focused local regression suite is green. This verifies the tested prototype paths; it is not a production readiness or live candidate-data assessment.